E-Commerce Database Management System

Week 5: Payment Transaction Management System

1. Payment Table Design

Week 5 completes the payment side of the order process. The payment table created in Task 1 stores the payment made for an order. This week it is linked to every order, payments are recorded in different payment modes, statuses are managed, and the payment data is analysed.

Week 5 objectives

- Use the payment table and the Orders to Payment relationship, and make sure each order has at most one payment record.
- Apply primary key, foreign key, NOT NULL, UNIQUE and CHECK constraints.
- Insert payments for orders in different payment modes: UPI, Credit Card, Debit Card, Net Banking and Cash on Delivery.
- Update payment statuses (Pending, Success, Failed), including a retry after a failed payment.
- Generate payment transactions and revenue reports.

How Week 5 builds on earlier tasks. The payment table from Task 1 already has the Week 5 columns, and payment.order_id already links each payment to its order. This week one more CHECK rule is added, and payments are inserted for the five new orders created in Week 4 (orders 11, 12, 13, 14 and 16). The amount of each payment is copied from the order total, so a payment can never disagree with its order.

Design rules used in this week

- An order has at most one payment record. A failed payment is retried by updating the same record, not by adding a second one.
- A pending payment has no payment date until the money is received.
- Revenue figures count only payments with the status Success.

1.1 Payment (table: payment)

Attribute	Data type	Constraint	Description
payment_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID of the payment
order_id	INT	NOT NULL, UNIQUE, FOREIGN KEY references orders(order_id), ON DELETE CASCADE	Order that is being paid for

Attribute	Data type	Constraint	Description
payment_date	DATE	Optional, required when the status is Success	Date of the payment (empty until the payment happens)
payment_mode	ENUM	NOT NULL	UPI, Credit Card, Debit Card, Net Banking or Cash on Delivery
payment_status	ENUM	NOT NULL, DEFAULT 'Pending'	Pending, Success or Failed
transaction_amount	DECIMAL(10,2)	NOT NULL, CHECK (transaction_amount >= 0)	Amount of the transaction

1.2 Payment Modes and Statuses

Field	Allowed values	Meaning
payment_mode	UPI	Instant payment from a bank account through an app
payment_mode	Credit Card	Payment by credit card
payment_mode	Debit Card	Payment by debit card
payment_mode	Net Banking	Payment through the customer's online banking
payment_mode	Cash on Delivery	Cash paid when the order is delivered
payment_status	Pending	The payment has not been received yet
payment_status	Success	The payment has been received
payment_status	Failed	The transaction did not go through

Both fields are ENUM columns, so MySQL accepts only these values. Anything else, such as Bitcoin as a payment mode, is rejected.

1.3 Constraint Summary

Constraint	Purpose	Where it is used
PRIMARY KEY	Identifies each payment uniquely	payment.payment_id
FOREIGN KEY	Links a payment to an existing order	payment.order_id to orders

Constraint	Purpose	Where it is used
NOT NULL	Makes a value compulsory	order_id, payment_mode, payment_status, transaction_amount
UNIQUE	Allows at most one payment per order	payment.order_id
CHECK	Rejects invalid values	transaction_amount >= 0, and a Success payment must have a payment_date
DEFAULT	Supplies a value when none is given	payment_status ('Pending')
ENUM	Limits a column to a fixed set of values	payment_mode, payment_status

1.4 Relationship Between Orders and Payment

Relationship	Type	Linked by	Explanation
Orders to Payment	One-to-One	payment.order_id (UNIQUE)	Each order has at most one payment record, and each payment belongs to exactly one order.

Referential integrity rules

- A payment cannot be recorded for an order that does not exist (MySQL error 1452).
- A second payment cannot be added for an order that already has one (error 1062).
- Deleting an order also deletes its payment record (ON DELETE CASCADE).
- A payment with the status Success must have a payment date (error 3819).
- The payment mode must be one of the five allowed values (error 1265).

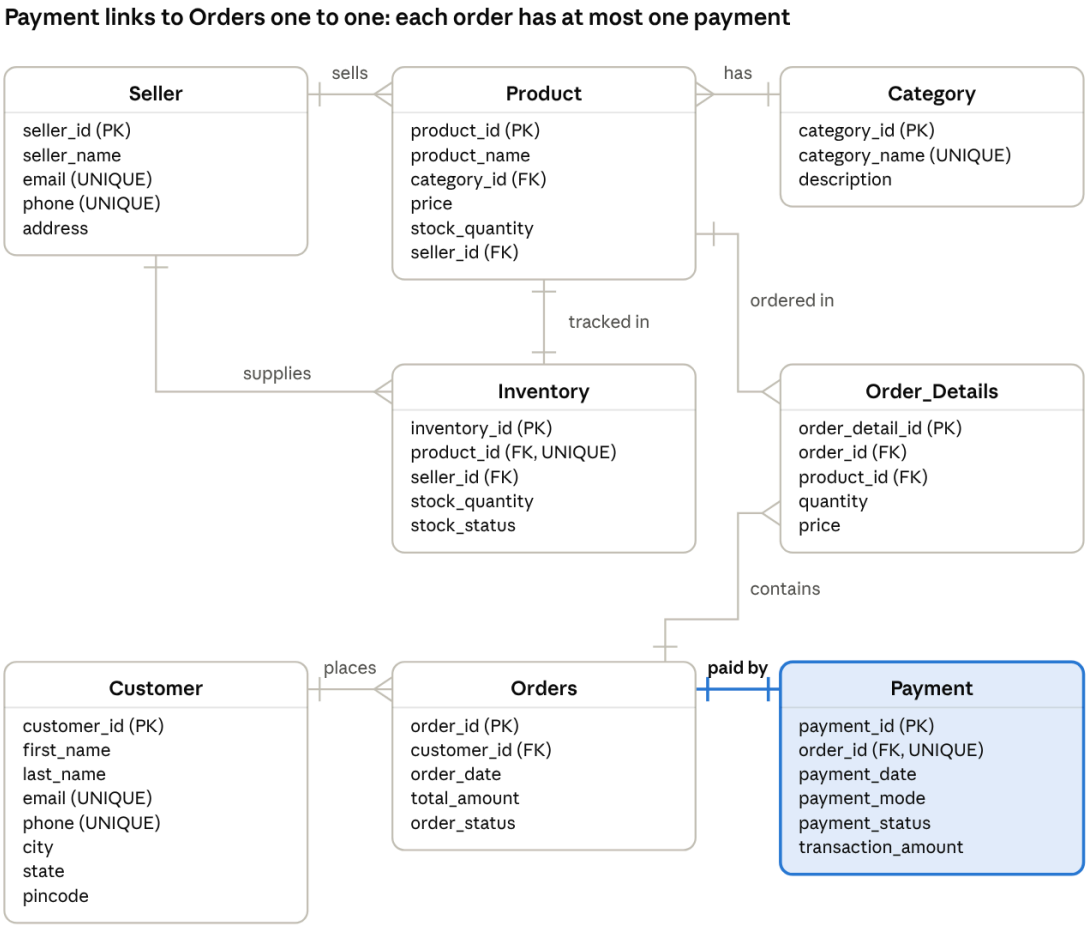
1.5 Keeping Payments Linked to Orders

Two checks confirm that the payment records stay linked to the Order Management System. After the payments are inserted, both queries return no rows.

- Orders without a payment record: every order has a payment.
- Payments whose amount differs from the order total: every transaction amount equals its order total.

2. ER Diagram

The updated ER diagram highlights the Order to Payment relationship. Each order has at most one payment record, which is enforced by the UNIQUE constraint on payment.order_id. The crow's foot marks the many side of a relationship.



Updated ER diagram · Orders to Payment relationship

A customer places many orders, and each order has at most one payment record. The relationships from Tasks 1 to 4 are unchanged.

3. Sample Customer, Order and Payment Data

3.1 Customer data

Customer_ID	Name	Email	City	State
1	Arun Kumar	arun@mail.com	Chennai	Tamil Nadu
2	Priya Sharma	priya@mail.com	Chennai	Tamil Nadu
3	Rahul Verma	rahul@mail.com	Mumbai	Maharashtra
4	Sneha Reddy	sneha@mail.com	Hyderabad	Telangana

Customer_ID	Name	Email	City	State
5	Karthik Raja	karthik@mail.com	Madurai	Tamil Nadu
6	Divya Nair	divya@mail.com	Kochi	Kerala
7	Vikram Singh	vikram@mail.com	Delhi	Delhi
8	Meena Iyer	meena@mail.com	Coimbatore	Tamil Nadu
9	Suresh Babu	suresh@mail.com	Bengaluru	Karnataka

3.2 Order data

These are the 15 orders at the start of Week 5, with the statuses left by Week 4. Order 15 was deleted in Week 4.

Order_ID	Customer	Order_Date	Total_Amount	Order_Status
1	Arun Kumar	2026-10-01	17999.00	Delivered
2	Priya Sharma	2026-10-02	998.00	Delivered
3	Rahul Verma	2026-10-03	1899.00	Shipped
4	Sneha Reddy	2026-10-04	750.00	Shipped
5	Karthik Raja	2026-10-05	1299.00	Delivered
6	Divya Nair	2026-10-06	1499.00	Pending
7	Vikram Singh	2026-10-07	3299.00	Delivered
8	Meena Iyer	2026-10-08	398.00	Cancelled
9	Suresh Babu	2026-10-09	799.00	Pending
10	Arun Kumar	2026-10-10	700.00	Pending
11	Rahul Verma	2026-10-11	5396.00	Delivered
12	Arun Kumar	2026-10-12	3697.00	Delivered
13	Priya Sharma	2026-10-13	2395.00	Shipped
14	Vikram Singh	2026-10-14	3998.00	Delivered
16	Divya Nair	2026-10-16	5198.00	Shipped

3.3 Payment data

Payments 1 to 10 come from Task 1. Payments 11 to 15 are inserted in Week 5 for the five new orders, using four different payment modes and all three statuses. Each amount is copied from the order total. A date shown as Empty means the payment has not been received yet.

Payment_ID	Order_ID	Customer	Payment_Date	Payment_Mode	Payment_Status	Transaction_Amount
1	1	Arun Kumar	2026-10-01	UPI	Success	17999.00
2	2	Priya Sharma	2026-10-02	Credit Card	Success	998.00
3	3	Rahul Verma	2026-10-03	Debit Card	Success	1899.00
4	4	Sneha Reddy	Empty	Cash on Delivery	Pending	750.00
5	5	Karthik Raja	2026-10-05	Net Banking	Success	1299.00
6	6	Divya Nair	2026-10-06	UPI	Success	1499.00
7	7	Vikram Singh	2026-10-07	Debit Card	Success	3299.00
8	8	Meena Iyer	2026-10-08	UPI	Failed	398.00
9	9	Suresh Babu	Empty	Cash on Delivery	Pending	799.00
10	10	Arun Kumar	2026-10-10	Credit Card	Success	700.00
11	11	Rahul Verma	2026-10-11	Net Banking	Success	5396.00
12	12	Arun Kumar	2026-10-12	UPI	Success	3697.00
13	13	Priya Sharma	2026-10-13	Credit Card	Failed	2395.00
14	14	Vikram Singh	2026-10-16	Cash on Delivery	Success	3998.00
15	16	Divya Nair	Empty	Debit Card	Pending	5198.00

3.4 Change made by the status update

Payment	Operation	Before	After
Payment 13 (order 13, Priya Sharma)	Retry after a failed transaction	Credit Card, Failed, 2026-10-13	UPI, Success, 2026-10-14

The retry updates the existing payment record, because an order can have only one payment. The update also sets the payment date, because a Success payment must have one. The analysis report below uses the data after this change.

4. Payment Transaction and Revenue Analysis Reports

This report presents the results of the payment queries in the SQL script, followed by the main observations.

4.1 Complete payment information

Payment_ ID	Customer	Order_I D	Order_St atus	Payment_ Mode	Payment_ Status	Payment_ Date	Amount
1	Arun Kumar	1	Delivered	UPI	Success	2026-10-01	17999.00
2	Priya Sharma	2	Delivered	Credit Card	Success	2026-10-02	998.00
3	Rahul Verma	3	Shipped	Debit Card	Success	2026-10-03	1899.00
4	Sneha Reddy	4	Shipped	Cash on Delivery	Pending	Empty	750.00
5	Karthik Raja	5	Delivered	Net Banking	Success	2026-10-05	1299.00
6	Divya Nair	6	Pending	UPI	Success	2026-10-06	1499.00
7	Vikram Singh	7	Delivered	Debit Card	Success	2026-10-07	3299.00
8	Meena Iyer	8	Cancelled	UPI	Failed	2026-10-08	398.00
9	Suresh Babu	9	Pending	Cash on Delivery	Pending	Empty	799.00

Payment_ ID	Customer	Order_I D	Order_St atus	Payment_ Mode	Payment_ Status	Payment_ Date	Amount
10	Arun Kumar	10	Pending	Credit Card	Success	2026-10-10	700.00
11	Rahul Verma	11	Delivered	Net Banking	Success	2026-10-11	5396.00
12	Arun Kumar	12	Delivered	UPI	Success	2026-10-12	3697.00
13	Priya Sharma	13	Shipped	UPI	Success	2026-10-14	2395.00
14	Vikram Singh	14	Delivered	Cash on Delivery	Success	2026-10-16	3998.00
15	Divya Nair	16	Shipped	Debit Card	Pending	Empty	5198.00

4.2 Successful, failed and pending payments

The query for successful payments runs before the retry of order 13, so it returns 10 rows at that point. After the retry there are 11 successful payments, 1 failed payment and 3 pending payments.

Payment_Status	Total_Transactions	Total_Amount
Success	11	43179.00
Pending	3	6747.00
Failed	1	398.00

The failed and pending transactions after the retry are:

Payment_ID	Order_ID	Payment_Mode	Payment_Status	Amount
8	8	UPI	Failed	398.00
4	4	Cash on Delivery	Pending	750.00
9	9	Cash on Delivery	Pending	799.00
15	16	Debit Card	Pending	5198.00

When the query first runs, before the retry, it also shows payment 13 (order 13, Credit Card, Failed), so it returns 5 rows.

4.3 Number of transactions by payment mode

Payment_Mode	Total_Transactions
UPI	5
Cash on Delivery	3
Debit Card	3
Credit Card	2
Net Banking	2

The most preferred payment method is **UPI**, with 5 of the 15 transactions.

4.4 Revenue from successful payments

The total successful payment revenue is **43179.00**. By payment method it is:

Payment_Mode	Successful_Payments	Revenue
UPI	4	25590.00
Net Banking	2	6695.00
Debit Card	2	5198.00
Cash on Delivery	1	3998.00
Credit Card	2	1698.00

4.5 Average transaction amount

Measure	Result
Average of all 15 transactions	3354.93
Average of the 11 successful transactions	3925.36

4.6 Customer payment history

Customer	Order_ID	Payment_ID	Payment_Mode	Payment_Status	Amount	Payment_Date
Arun Kumar	1	1	UPI	Success	17999.00	2026-10-01
Arun Kumar	10	10	Credit Card	Success	700.00	2026-10-10
Arun Kumar	12	12	UPI	Success	3697.00	2026-10-12
Priya Sharma	2	2	Credit Card	Success	998.00	2026-10-02

Customer	Order_ID	Payment_ID	Payment_Mode	Payment_Status	Amount	Payment_Date
Priya Sharma	13	13	UPI	Success	2395.00	2026-10-14
Rahul Verma	3	3	Debit Card	Success	1899.00	2026-10-03
Rahul Verma	11	11	Net Banking	Success	5396.00	2026-10-11
Sneha Reddy	4	4	Cash on Delivery	Pending	750.00	Empty
Karthik Raja	5	5	Net Banking	Success	1299.00	2026-10-05
Divya Nair	6	6	UPI	Success	1499.00	2026-10-06
Divya Nair	16	15	Debit Card	Pending	5198.00	Empty
Vikram Singh	7	7	Debit Card	Success	3299.00	2026-10-07
Vikram Singh	14	14	Cash on Delivery	Success	3998.00	2026-10-16
Meena Iyer	8	8	UPI	Failed	398.00	2026-10-08
Suresh Babu	9	9	Cash on Delivery	Pending	799.00	Empty

4.7 Orders with failed or pending payments

Order_ID	Customer	Order_Status	Payment_Mode	Payment_Status	Amount
4	Sneha Reddy	Shipped	Cash on Delivery	Pending	750.00
8	Meena Iyer	Cancelled	UPI	Failed	398.00
9	Suresh Babu	Pending	Cash on Delivery	Pending	799.00
16	Divya Nair	Shipped	Debit Card	Pending	5198.00

4.8 Key observations

- UPI is the most preferred payment method. It accounts for 5 of the 15 transactions and 25590.00, about 59%, of the successful revenue.
- 11 of the 15 payments (about 73%) are successful. 3 are pending and 1 has failed.
- 6747.00 is still waiting to be collected. Of this, 5198.00 belongs to order 16, which is paid by debit card and already shipped.

- Cash on Delivery has 3 transactions but only 1 is successful. The other 2 stay pending until delivery, so it carries the most collection risk.
- The retry of order 13 recovered 2395.00 that would otherwise have been lost after the failed credit card payment.
- Orders 6 and 10 are fully paid but are still Pending, so they can be shipped.
- The only failed payment belongs to the cancelled order 8, so no active order has a failed payment.

5. Query Output Screenshots

1. DESCRIBE payment

Field	Type	Null	Key	Default	Extra
payment_id	int	NO	PRI	NULL	auto_increment
order_id	int	NO	UNI	NULL	
payment_date	date	YES		NULL	
payment_mode	enum('UPI','Credit Card','Debit Card','Net Bankin...	NO		NULL	
payment_status	enum('Pending','Success','Failed')	NO		Pending	
transaction_amount	decimal(10,2)	NO		NULL	

2. SHOW COLUMNS for payment_mode and payment_status

Field	Type	Null	Key	Default	Extra
payment_mode	enum('UPI','Credit Card','Debit Card','Net Bankin...	NO		NULL	

Field	Type	Null	Key	Default	Extra
payment_status	enum('Pending','Success','Failed')	NO		Pending	

3. SELECT * FROM payment

payment_id	order_id	payment_date	payment_mode	payment_status	transaction_amount
1	1	2026-10-01	UPI	Success	17999.00
2	2	2026-10-02	Credit Card	Success	998.00
3	3	2026-10-03	Debit Card	Success	1899.00
4	4	NULL	Cash on Delivery	Pending	750.00
5	5	2026-10-05	Net Banking	Success	1299.00
6	6	2026-10-06	UPI	Success	1499.00
7	7	2026-10-07	Debit Card	Success	3299.00
8	8	2026-10-08	UPI	Failed	398.00
9	9	NULL	Cash on Delivery	Pending	799.00
10	10	2026-10-10	Credit Card	Success	700.00
11	11	2026-10-11	Net Banking	Success	5396.00
12	12	2026-10-12	UPI	Success	3697.00
13	13	2026-10-14	UPI	Success	2395.00
14	14	2026-10-16	Cash on Delivery	Success	3998.00
15	16	NULL	Debit Card	Pending	5198.00
NULL	NULL	NULL	NULL	NULL	NULL

4. Orders without a payment

order_id	customer_id	order_date	total_amount	order_status

5. SELECT * FROM payment WHERE order_id = 13

payment_id	order_id	payment_date	payment_mode	payment_status	transaction_amount
13	13	2026-10-14	UPI	Success	2395.00
NULL	NULL	NULL	NULL	NULL	NULL

6. The two link checks

order_id

payment_id	order_id	transaction_amount	total_amount

7. Successful payments

payment_id	order_id	payment_date	payment_mode	transaction_amount
1	1	2026-10-01	UPI	17999.00
2	2	2026-10-02	Credit Card	998.00
3	3	2026-10-03	Debit Card	1899.00
5	5	2026-10-05	Net Banking	1299.00
6	6	2026-10-06	UPI	1499.00
7	7	2026-10-07	Debit Card	3299.00
10	10	2026-10-10	Credit Card	700.00
11	11	2026-10-11	Net Banking	5396.00
12	12	2026-10-12	UPI	3697.00
13	13	2026-10-14	UPI	2395.00
14	14	2026-10-16	Cash on Delivery	3998.00
NULL	NULL	NULL	NULL	NULL

8. Failed and pending payments

payment_id	order_id	payment_mode	payment_status	transaction_amount
4	4	Cash on Delivery	Pending	750.00
9	9	Cash on Delivery	Pending	799.00
15	16	Debit Card	Pending	5198.00
8	8	UPI	Failed	398.00
NULL	NULL	NULL	NULL	NULL

9. Count of successful and failed payments

successful_payments	failed_payments
11	1

10. Complete payment information

payment_id	customer_name	order_id	order_date	order_status	total_amount	payment_mode	payment_status	payment_date	transaction_amount
1	Arun Kumar	1	2026-10-01	Delivered	17999.00	UPI	Success	2026-10-01	17999.00
2	Priya Sharma	2	2026-10-02	Delivered	998.00	Credit Card	Success	2026-10-02	998.00
3	Rahul Verma	3	2026-10-03	Shipped	1899.00	Debit Card	Success	2026-10-03	1899.00
4	Sneha Reddy	4	2026-10-04	Shipped	750.00	Cash on Delivery	Pending	NULL	750.00
5	Karthik Raja	5	2026-10-05	Delivered	1299.00	Net Banking	Success	2026-10-05	1299.00
6	Divya Nair	6	2026-10-06	Pending	1499.00	UPI	Success	2026-10-06	1499.00
7	Vikram Singh	7	2026-10-07	Delivered	3299.00	Debit Card	Success	2026-10-07	3299.00
8	Meena Iyer	8	2026-10-08	Cancelled	398.00	UPI	Failed	2026-10-08	398.00
9	Suresh Babu	9	2026-10-09	Pending	799.00	Cash on Delivery	Pending	NULL	799.00
10	Arun Kumar	10	2026-10-10	Pending	700.00	Credit Card	Success	2026-10-10	700.00
11	Rahul Verma	11	2026-10-11	Delivered	5396.00	Net Banking	Success	2026-10-11	5396.00
12	Arun Kumar	12	2026-10-12	Delivered	3697.00	UPI	Success	2026-10-12	3697.00
13	Priya Sharma	13	2026-10-13	Shipped	2395.00	UPI	Success	2026-10-14	2395.00
14	Vikram Singh	14	2026-10-14	Delivered	3998.00	Cash on Delivery	Success	2026-10-16	3998.00
15	Divya Nair	16	2026-10-16	Shipped	5198.00	Debit Card	Pending	NULL	5198.00

11. Transactions by payment mode

payment_mode	total_transactions
UPI	5
Cash on Delivery	3
Debit Card	3
Credit Card	2
Net Banking	2

12. Most preferred payment method

payment_mode	total_transactions
UPI	5

13. Total successful revenue

total_successful_revenue
43179.00

14. Revenue by payment method

payment_mode	successful_payments	revenue
UPI	4	25590.00
Net Banking	2	6695.00
Debit Card	2	5198.00
Cash on Delivery	1	3998.00
Credit Card	2	1698.00

15. Average transaction amount

average_all_transactions
3354.93

average_successful_transactions
3925.36

16. Customer payment history

customer_id	customer_name	order_id	payment_id	payment_mode	payment_status	transaction_amount	payment_date
1	Arun Kumar	1	1	UPI	Success	17999.00	2026-10-01
1	Arun Kumar	10	10	Credit Card	Success	700.00	2026-10-10
1	Arun Kumar	12	12	UPI	Success	3697.00	2026-10-12
2	Priya Sharma	2	2	Credit Card	Success	998.00	2026-10-02
2	Priya Sharma	13	13	UPI	Success	2395.00	2026-10-14
3	Rahul Verma	3	3	Debit Card	Success	1899.00	2026-10-03
3	Rahul Verma	11	11	Net Banking	Success	5396.00	2026-10-11
4	Sneha Reddy	4	4	Cash on Delivery	Pending	750.00	NULL
5	Karthik Raja	5	5	Net Banking	Success	1299.00	2026-10-05
6	Divya Nair	6	6	UPI	Success	1499.00	2026-10-06
6	Divya Nair	16	15	Debit Card	Pending	5198.00	NULL
7	Vikram Singh	7	7	Debit Card	Success	3299.00	2026-10-07
7	Vikram Singh	14	14	Cash on Delivery	Success	3998.00	2026-10-16
8	Meena Iyer	8	8	UPI	Failed	398.00	2026-10-08
9	Suresh Babu	9	9	Cash on Delivery	Pending	799.00	NULL

17. Successful, failed and pending totals

payment_status	total_transactions	total_amount
Success	11	43179.00
Pending	3	6747.00
Failed	1	398.00

18. Orders with failed or pending payments

order_id	customer_name	order_status	payment_mode	payment_status	transaction_amount
4	Sneha Reddy	Shipped	Cash on Delivery	Pending	750.00
8	Meena Iyer	Cancelled	UPI	Failed	398.00
9	Suresh Babu	Pending	Cash on Delivery	Pending	799.00
16	Divya Nair	Shipped	Debit Card	Pending	5198.00

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II BCA Batch 1